Software pros, listen up! Labour day Offer ends tonight, you can get 25% off on enrolling in 2 or more courses at once. Use Code: SAVINGS25

Practice Set 2

Your results are here!! for" Databricks Certified Data Engineer Associate Practice Test 2 "

0 of 65 questions answered correctly

Your time: 00:00:32

Your Final Score is : 0

You have attempted : 0

Number of Correct Questions : 0 and scored 0

Number of Incorrect Questions : 0 and Negative marks 0

Average score	57.69%
Your score	0.00%

You can review your answers by clicking on “View Answers” option.
Important Note : Open Reference Documentation Links in New Tab (Right Click and Open in New Tab).

Restart Test View Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64

65

Answered Review

1. Question

The data analyst team had put together queries that identify items that are out of stock based on orders and replenishment but when they run all together for final output the team noticed it takes a really long time, you were asked to look at the reason why queries are running slow and identify steps to improve the performance and when you looked at it you noticed all the code queries are running sequentially and using a SQL endpoint cluster. Which of the following steps can be taken to resolve the issue?

Here is the example query

```
-- Get order summary
create or replace table orders_summary
as
select product_id, sum(order_count) order_count
from
(
select product_id,order_count from orders_instore
union all
select product_id,order_count from orders_online
)
group by product_id
-- get supply summary
create or repalce tabe supply_summary
as
select product_id, sum(supply_count) supply_count
from supply
group by product_id
-- get on hand based on orders summary and supply summary
with stock_cte
as (
select nvl(s.product_id,o.product_id) as product_id,
nvl(supply_count,0) - nvl(order_count,0) as on_hand
from supply_summary s
full outer join orders_summary o
on s.product_id = o.product_id
)
select *
from
stock_cte
where on_hand = 0
```

- Turn on the Serverless feature for the SQL endpoint.

- Increase the maximum bound of the SQL endpoint's scaling range.
- **Increase the cluster size of the SQL endpoint.**
- Turn on the Auto Stop feature for the SQL endpoint.
- Turn on the Serverless feature for the SQL endpoint and change the Spot Instance Policy to "Reliability Optimized."

Unattempted

The answer is to increase the cluster size of the SQL Endpoint, here queries are running sequentially and since the single query can not span more than one cluster adding more clusters won't improve the query but rather increasing the cluster size will improve performance so it can use additional compute in a warehouse.

In the exam please note that additional context will not be given instead you have to look for cue words or need to understand if the queries are running sequentially or concurrently. if the queries are running sequentially then scale up(more nodes) if the queries are running concurrently (more users) then scale out(more clusters).

Below is the snippet from Azure, as you can see by increasing the cluster size you are able to add more worker nodes.

Cluster size	Instance type for driver(applies only to Classic Warehouses/SQL Endpoints)	Worker count
2X-Small	Standard_E8ds_v4	1
X-Small		2
Small	Standard_E16ds_v4	4
Medium	Standard_E32ds_v4	8
Large		16
X-Large	Standard_E64ds_v4	32
2X-Large		64
3X-Large		128
4X-Large		256

*The instance size of all workers is Standard_E8ds_v4.

SQL endpoint scales horizontally(scale-out) and vertically (scale-up), you have to understand when to use what.

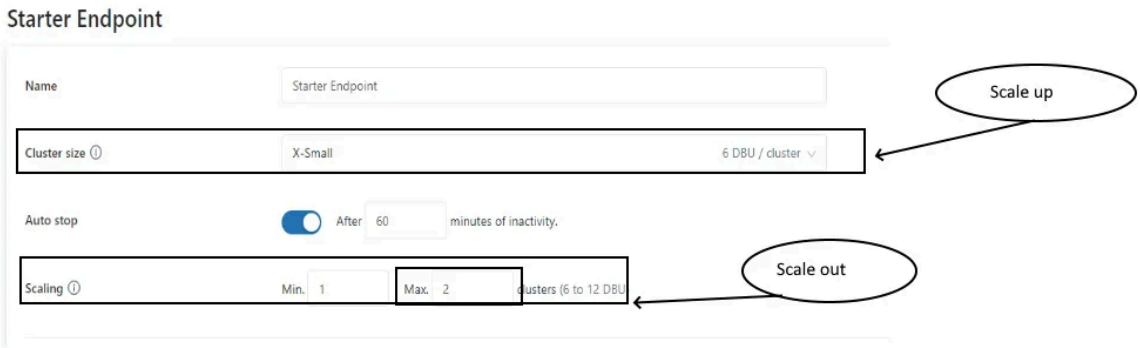
Scale-up-> Increase the size of the cluster from x-small to small, to medium, X Large....

If you are trying to improve the performance of a single query having additional memory, additional nodes and cpu in the cluster will improve the performance.

Scale-out -> Add more clusters, change max number of clusters

If you are trying to improve the throughput, being able to run as many queries as possible then having an additional cluster(s) will improve the performance.

SQL endpoint



2. Question

The operations team is interested in monitoring the recently launched product, team wants to set up an email alert when the number of units sold increases by more than 10,000 units. They want to monitor this every 5 mins.

Fill in the below blanks to finish the steps we need to take

· Create ___ query that calculates total units sold

· Setup ____ with query on trigger condition Units Sold > 10,000

· Setup ____ to run every 5 mins

· Add destination _____

- Python, Job, SQL Cluster, email address
- **SQL, Alert, Refresh, email address**
- SQL, Job, SQL Cluster, email address
- SQL, Job, Refresh, email address
- Python, Job, Refresh, email address

Unattempted

The answer is SQL, Alert, Refresh, email address

Here the steps from Databricks documentation,

Create an alert

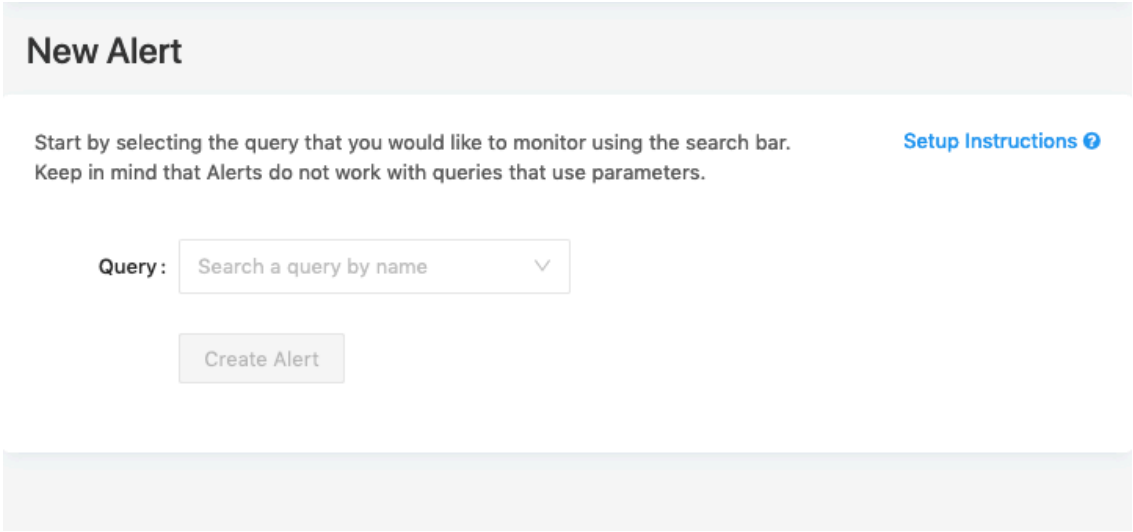
Follow these steps to create an alert on a single column of a query.

1. Do one of the following:

Click Create in the sidebar and select Alert.

Click Alerts in the sidebar and click the + New Alert button.

2. Search for a target query.



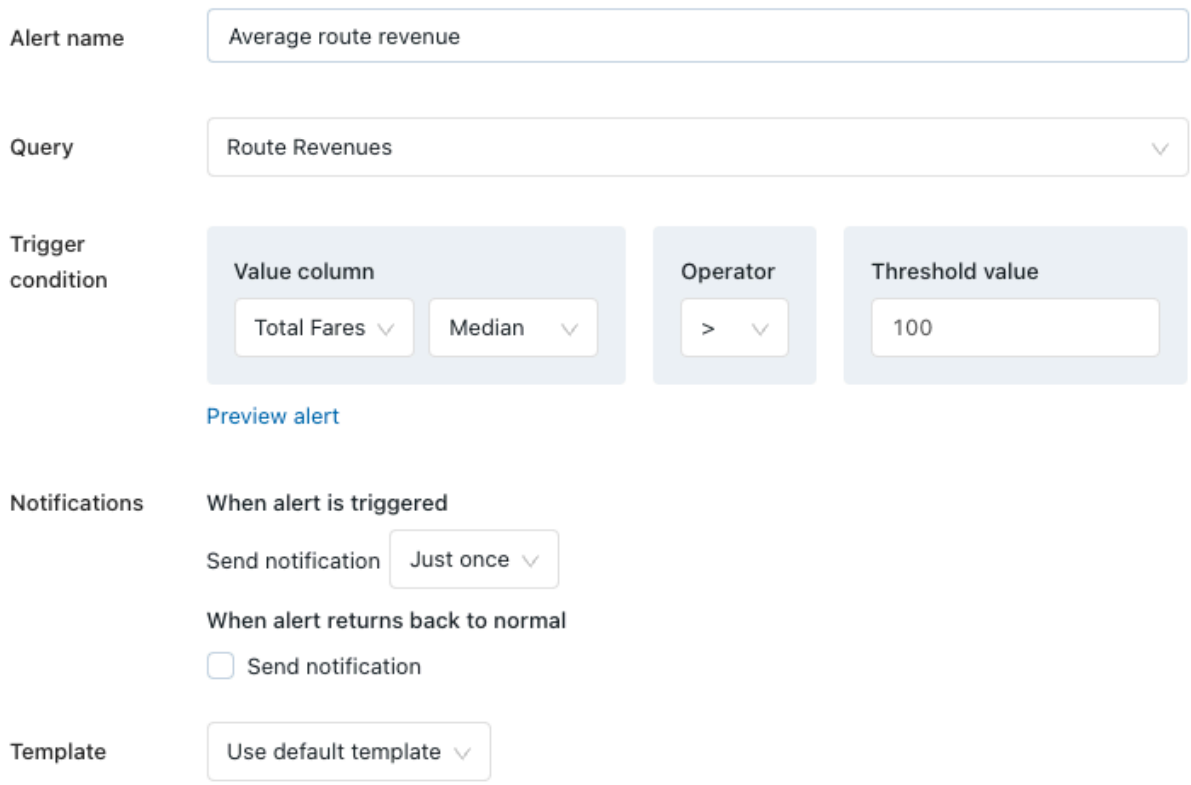
To alert on multiple columns, you need to modify your query. See Alert on multiple columns.

3. In the Trigger when field, configure the alert.

The Value column drop-down controls which field of your query result is evaluated.

The Condition drop-down controls the logical operation to be applied.

The Threshold text input is compared against the Value column using the Condition you specify.



Note

If a target query returns multiple records, Databricks SQL alerts act on the first one. As you change the Value column setting, the current value of that field in the top row is shown beneath it.

4. In the When triggered, send notification field, select how many notifications are sent when your alert is triggered:

Just once: Send a notification when the alert status changes from OK to TRIGGERED.

Each time alert is evaluated: Send a notification whenever the alert status is TRIGGERED regardless of its status at the previous evaluation.

At most every: Send a notification whenever the alert status is TRIGGERED at a specific interval. This choice lets you avoid notification spam for alerts that trigger often.

Regardless of which notification setting you choose, you receive a notification whenever the status goes from OK to TRIGGERED or from TRIGGERED to OK. The schedule settings affect how many notifications you will receive if the status remains TRIGGERED from one execution to the next. For details, see Notification frequency.

5. In the Template drop-down, choose a template:

Use default template: Alert notification is a message with links to the Alert configuration screen and the Query screen.

Use custom template: Alert notification includes more specific information about the alert.

a. A box displays, consisting of input fields for subject and body. Any static content is valid, and you can incorporate built-in template variables:

ALERT_STATUS: The evaluated alert status (string).

ALERT_CONDITION: The alert condition operator (string).

ALERT_THRESHOLD: The alert threshold (string or number).

ALERT_NAME: The alert name (string).

ALERT_URL: The alert page URL (string).

QUERY_NAME: The associated query name (string).

QUERY_URL: The associated query page URL (string).

QUERY_RESULT_VALUE: The query result value (string or number).

QUERY_RESULT_ROWS: The query result rows (value array).

QUERY_RESULT_COLS: The query result columns (string array).

An example subject, for instance, could be: Alert “{{ALERT_NAME}}“ changed status to {{ALERT_STATUS}}.

b. Click the Preview toggle button to preview the rendered result.

Important

The preview is useful for verifying that template variables are rendered correctly. It is not an accurate representation of the eventual notification content, as each alert destination can display notifications differently.

c. Click the Save Changes button.

6. In Refresh, set a refresh schedule. An alert’s refresh schedule is independent of the query’s refresh schedule.

If the query is a Run as owner query, the query runs using the query owner’s credential on the alert’s refresh schedule.

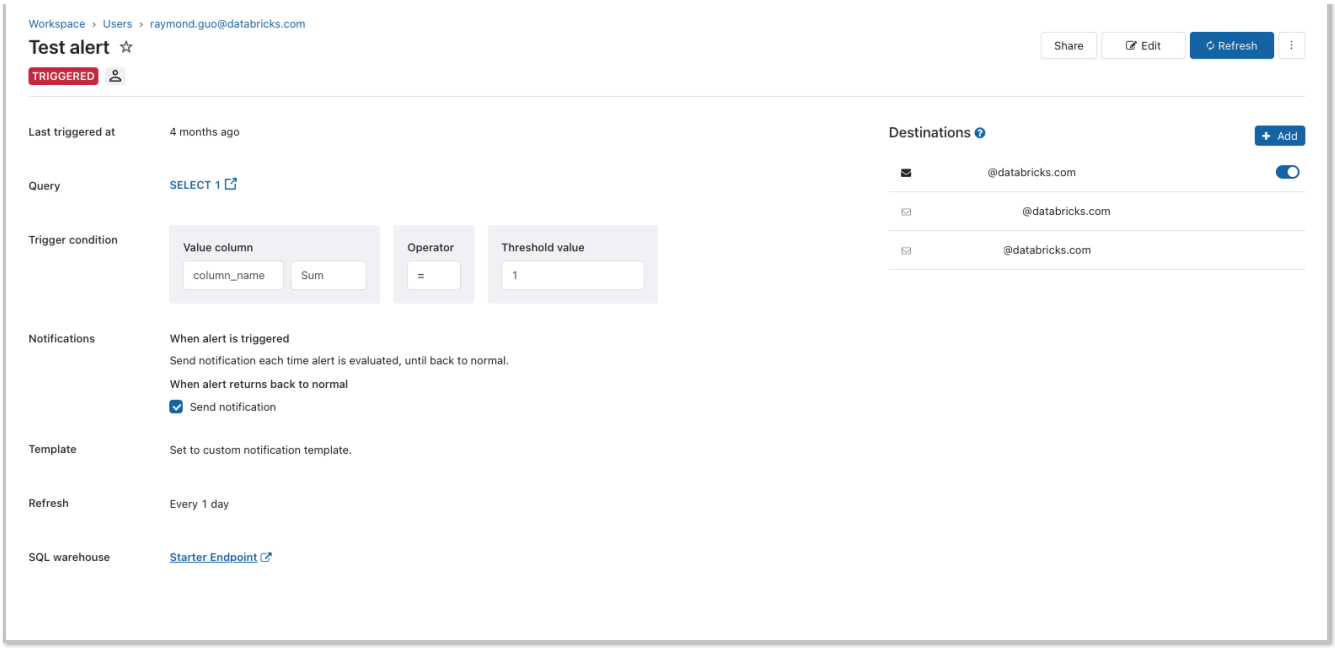
If the query is a Run as viewer query, the query runs using the alert creator’s credential on the alert’s refresh schedule.

7. Click Create Alert.

8. Choose an alert destination.

Important

If you skip this step you will not be notified when the alert is triggered.



3. Question

The marketing team is launching a new campaign to monitor the performance of the new campaign for the first two weeks, they would like to set up a dashboard with a refresh schedule to run every 5 minutes, which of the below steps can be taken to reduce of the cost of this refresh over time?

- Reduce the size of the SQL Cluster size
- Reduce the max size of auto scaling from 10 to 5
- **Setup the dashboard refresh schedule to end in two weeks**
- Change the spot instance policy from reliability optimized to cost optimized
- Always use X-small cluster

Unattempted

The answer is Setup the dashboard refresh schedule to end in two weeks

4. Question

Which of the following tool provides Data Access control, Access Audit, Data Lineage, and Data discovery?

- DELTA LIVE Pipelines
- **Unity Catalog**
- Data Governance
- DELTA lake
- Lakehouse

Unattempted

The answer is Unity Catalog

5. Question

Data engineering team is required to share the data with Data science team and both the teams are using different workspaces in the same organizationwhich of the following techniques can be used to simplify sharing data across?

*Please note the question is asking how data is shared within an organization across multiple workspaces.

- Data Sharing
- Unity Catalog
- DELTA lake
- Use a single storage location
- DELTA LIVE Pipelines

Unattempted

The answer is the Unity catalog.

.

Unity Catalog works at the Account level, it has the ability to create a meta store and attach that meta store to many workspaces

see the below diagram to understand how Unity Catalog Works, as you can see a metastore can now be shared with both workspaces using Unity Catalog, prior to Unity Catalog the options was to use single cloud object storage manually mount in the second databricks workspace, and you can see here Unity Catalog really simplifies that.

Review product features
https://databricks.com/product/unity-catalog

6. Question

A newly joined team member John Smith in the Marketing team who currently does not have any access to the data requires read access to customers table, which of the following statements can be used to grant access.

- GRANT SELECT, USAGE TO john.smith@marketing.com ON TABLE customers
- GRANT READ, USAGE TO john.smith@marketing.com ON TABLE customers
- **GRANT SELECT, USAGE ON TABLE customers TO john.smith@marketing.com**
- GRANT READ, USAGE ON TABLE customers TO john.smith@marketing.com
- GRANT READ, USAGE ON customers TO john.smith@marketing.com

Unattempted

The answer is GRANT SELECT, USAGE ON TABLE customers TO john.smith@marketing.com
Data object privileges – Azure Databricks | Microsoft Docs

7. Question

Grant full privileges to new marketing user Kevin Smith to table sales

- GRANT FULL PRIVILEGES TO kevin.smith@marketing.com ON TABLE sales
- GRANT ALL PRIVILEGES TO kevin.smith@marketing.com ON TABLE sales
- GRANT FULL PRIVILEGES ON TABLE sales TO kevin.smith@marketing.com

- GRANT ALL PRIVILEGES ON TABLE sales TO kevin.smith@marketing.com
- GRANT ANY PRIVILEGE ON TABLE sales TO kevin.smith@marketing.com

Unattempted

The answer is GRANT ALL PRIVILEGE ON TABLE sales TO kevin.smith@marketing.com

GRANT ON TO Here are the available privileges and ALL Privileges gives full access to an object.

Privileges

SELECT: gives read access to an object.

CREATE: gives ability to create an object (for example, a table in a schema).

MODIFY: gives ability to add, delete, and modify data to or from an object.

USAGE: does not give any abilities, but is an additional requirement to perform any action on a schema object.

READ_METADATA: gives ability to view an object and its metadata.

CREATE_NAMED_FUNCTION: gives ability to create a named UDF in an existing catalog or schema.

MODIFY_CLASSPATH: gives ability to add files to the Spark class path.

ALL PRIVILEGES: gives all privileges (is translated into all the above privileges).

8. Question

Which of the following locations in the Databricks product architecture hosts the notebooks and jobs?

- Data plane
- Control plane
- Databricks Filesystem
- JDBC data source
- Databricks web application

Unattempted

The answer is Control Pane,

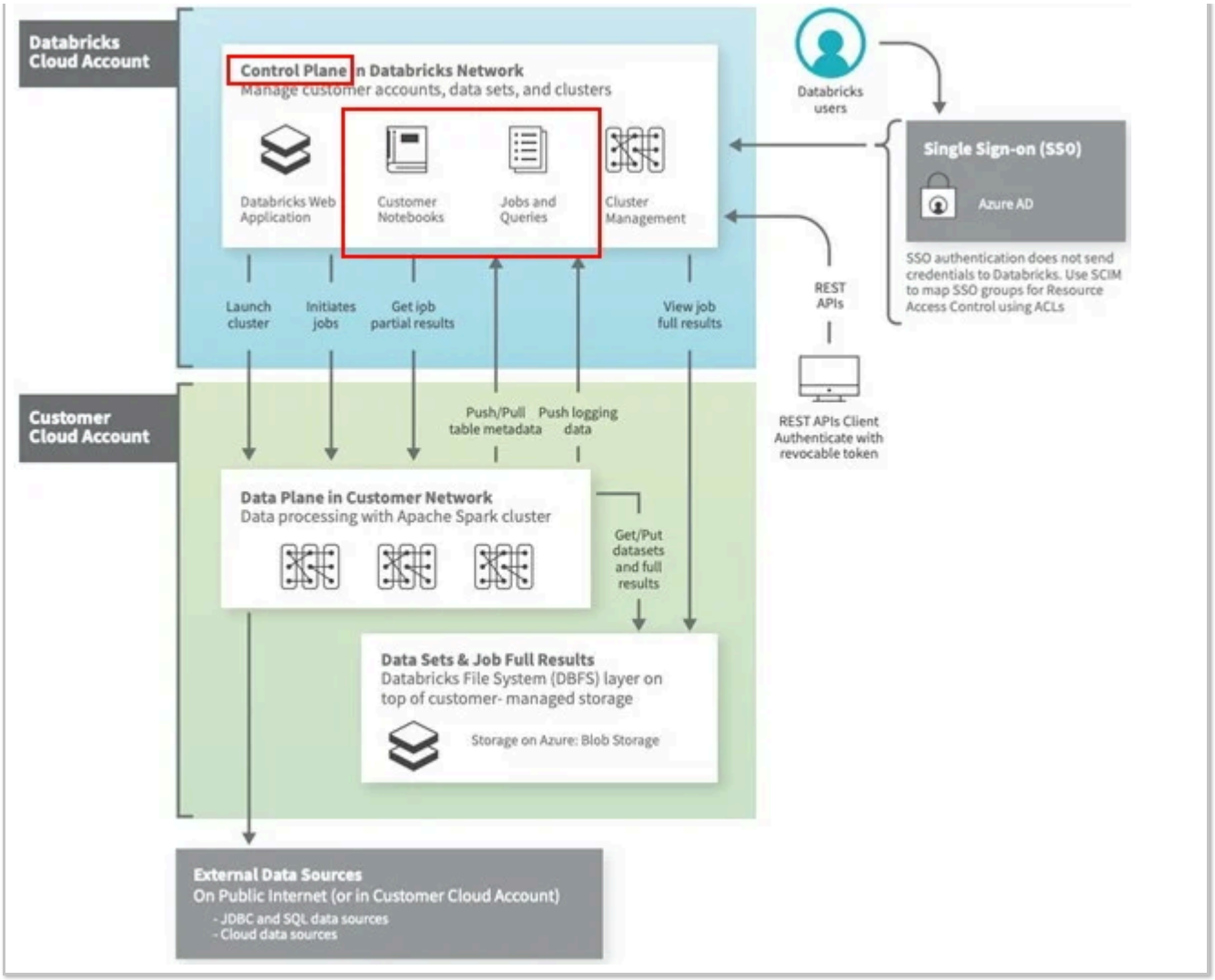
Databricks operates most of its services out of a control plane and a data plane, please note serverless features like SQL Endpoint and DLT compute use shared compute in Control pane.

Control Plane: Stored in Databricks Cloud Account

The control plane includes the backend services that Databricks manages in its own Azure account. Notebook commands and many other workspace configurations are stored in the control plane and encrypted at rest.

Data Plane: Stored in Customer Cloud Account

The data plane is managed by your Azure account and is where your data resides. This is also where data is processed. You can use Azure Databricks connectors so that your clusters can connect to external data sources outside of your Azure account to ingest data or for storage.



9. Question

A dataset has been defined using Delta Live Tables and includes an expectations clause: CONSTRAINT valid_timestamp EXPECT (timestamp > '2020-01-01') ON VIOLATION FAIL

What is the expected behavior when a batch of data containing data that violates these constraints is processed?

- Records that violate the expectation are added to the target dataset and recorded as invalid in the event log.
- Records that violate the expectation are dropped from the target dataset and recorded as invalid in the event log.
- **Records that violate the expectation cause the job to fail**
- Records that violate the expectation are added to the target dataset and flagged as invalid in a field added to the target dataset.
- Records that violate the expectation are dropped from the target dataset and loaded into a quarantine table.

Unattempted

The answer is Records that violate the expectation cause the job to fail.

Delta live tables support three types of expectations to fix bad data in DLT pipelines

Review below example code to examine these expectations,

Invalid records:
Use the expect operator when you want to keep records that violate the expectation. Records that violate the expectation are added to the target dataset along with valid records:
SQL
CONSTRAINT valid_timestamp EXPECT (timestamp > '2020-01-01')
Drop invalid records:
Use the expect or drop operator to prevent the processing of invalid records. Records that violate the expectation are dropped from the target dataset:
SQL
CONSTRAINT valid_timestamp EXPECT (timestamp > '2020-01-01') ON VIOLATION DROP ROW
Fail on invalid records:
When invalid records are unacceptable, use the expect or fail operator to halt execution immediately when a record fails validation. If the operation is a table update, the system atomically rolls back the transaction:
SQL
CONSTRAINT valid_timestamp EXPECT (timestamp > '2020-01-01') ON VIOLATION FAIL UPDATE

10. Question

You are still noticing slowness in query after performing optimize which helped you to resolve the small files problem, the column(transactionId) you are using to filter the data has high cardinality and auto incrementing number. Which delta optimization can you enable to filter data effectively based on this column?

- Create BLOOM FLTER index on the transactionId
- Perform Optimize with Zorder on transactionId
- transactionId has high cardinality, you cannot enable any optimization.

- Increase the cluster size and enable delta optimization
- Increase the driver size and enable delta optimization

Unattempted

The answer is, perform Optimize with Z-order by transactionid

Here is a simple explanation of how Z-order works, once the data is naturally ordered, when a flle is scanned it only brings the data it needs into spark‘s memory

Based on the column min and max it knows which data files needs to be scanned.

.

11. Question

If you create a database sample_db with the statement CREATE DATABASE sample_db what will be the default location of the database in DBFS?

- Default location, DBFS:/user/
- Default location, /user/db/

- Default Storage account
- Statement fails “Unable to create database without location”
- **Default Location, dbfs:/user/hive/warehouse**

Unattempted

The Answer is dbfs:/user/hive/warehouse this is the default location where spark stores user databases, the default can be changed using spark.sql.warehouse.dir a parameter. You can also provide a custom location using the LOCATION keyword.

Here is how this works,

.

Default location

FYI, This can be changed used using cluster spark config or session config.

Modify spark.sql.warehouse.dir location to change the default location

12. Question

Which of the following results in the creation of an external table?

- CREATE TABLE transactions (id int, desc string) USING DELTA LOCATION EXTERNAL
- CREATE TABLE transactions (id int, desc string)
- CREATE EXTERNAL TABLE transactions (id int, desc string)
- CREATE TABLE transactions (id int, desc string) TYPE EXTERNAL
- **CREATE TABLE transactions (id int, desc string) LOCATION '/mnt/delta/transactions'**

Unattempted

Answer is CREATE TABLE transactions (id int, desc string) USING DELTA LOCATION '/mnt/delta/transactions'

Anytime a table is created using Location it is considered an external table, below is the current syntax.
Syntax
CREATE TABLE table_name (column column_data_type...) USING format LOCATION “dbfs:/”

13. Question

When you drop an external DELTA table using the SQL Command DROP TABLE table_name, how does it impact metadata(delta log, history), and data stored in the storage?

- Drops table from metastore, metadata(delta log, history)and data in storage
- Drops table from metastore, data but keeps metadata(delta log, history) in storage
- Drops table from metastore, metadata(delta log, history)but keeps the data in storage
- **Drops table from metastore, but keeps metadata(delta log, history)and data in storage**
- Drops table from metastore and data in storage but keeps metadata(delta log, history)

Unattempted
The answer is Drops table from metastore, but keeps metadata and data in storage.
When an external table is dropped, only the table definition is dropped from metastore everything including data and metadata(Delta transaction log, time travel history) remains in the storage. Delta log is considered as part of metadata because if you drop a column in a delta table(managed or external) the column is not physically removed from the parquet files rather it is recorded in the delta log. The delta log becomes a key metadata layer for a Delta table to work.
Please see the below image to compare the external delta table and managed delta table and how they differ in how they are created and what happens if you drop the table.

14. Question

Which of the following is a true statement about the global temporary view?

- A global temporary view is available only on the cluster it was created, when the cluster restarts global temporary view is automatically dropped.
- A global temporary view is available on all clusters for a given workspace
- A global temporary view persists even if the cluster is restarted
- A global temporary view is stored in a user database
- A global temporary view is automatically dropped after 7 days

Unattempted

The answer is, A global temporary view is available only on the cluster it was created.
Two types of temporary views can be created Session scoped and Global
A session scoped temporary view is only available with a spark session, so another notebook in the same cluster can not access it. if a notebook is detached and re attached the temporary view is lost.
A global temporary view is available to all the notebooks in the cluster, if a cluster restarts global temporary view is lost.

15. Question

You are trying to create an object by joining two tables that and it is accessible to data scientist's team, so it does not get dropped if the cluster restarts or if the notebook is detached. What type of object are you trying to create?

- Temporary view
- Global Temporary view
- Global Temporary view with cache option
- External view
- View

Unattempted

Answer is View, A view can be used to join multiple tables but also persist into meta stores so others can accesses it

16. Question

What is the best way to query external csv files located on DBFS Storage to inspect the data using SQL?

- SELECT * FROM 'dbfs:/location/csv_files/' FORMAT = 'CSV'
- SELECT CSV. * from 'dbfs:/location/csv_files/'
- SELECT * FROM CSV. 'dbfs:/location/csv_files/'
- You can not query external files directly, us COPY INTO to load the data into a table first
- SELECT * FROM 'dbfs:/location/csv_files/' USING CSV

Unattempted

Answer is, SELECT * FROM CSV. 'dbfs:/location/csv_files/'

you can query external files stored on the storage using below syntax

SELECT * FROM format.'/Location'

format – CSV, JSON, PARQUET, TEXT

17. Question

Direct query on external files limited options, create external tables for CSV files with header and pipe delimited CSV files, fill in the blanks to complete the create table statement

CREATE TABLE sales (id int, unitsSold int, price FLOAT, items STRING)

_____

_____

LOCATION "dbfs:/mnt/sales/*.csv"

- FORMAT CSV
  OPTIONS ("true","|")
- USING CSV
  TYPE ("true","|")
- **USING CSV**
  OPTIONS (header ="true", delimiter = "|")
- FORMAT CSV
  FORMAT TYPE (header ="true", delimiter = "|")
- FORMAT CSV
  TYPE (header ="true", delimiter = "|")

Unattempted

USING CSV

OPTIONS (header ="true", delimiter = "|")

Here is the syntax to create an external table with additional options

CREATE TABLE table_name (col_name1 col_typ1,..)

USING data_source

OPTIONS (key='value', key2=vla2)

LOCATION = "/location"

18. Question

What could be the expected output of query SELECT COUNT (DISTINCT *) FROM user on this table

userId	username	email
1	john.smith	john.smith@com
2	NULL	david.clear@com
3	kevin.smith	kevin.smith@com

- 3
- 2
- 1
- NULL

Unattempted

Count(DISTINCT *) removes rows with any column with a NULL value

19. Question

You are working on a table called orders which contains data for 2021 and you have the second table called orders_archive which contains data for 2020, you need to combine the data from two tables and there could be a possibility of the same rows between both the tables, you are looking to combine the results from both the tables and eliminate the duplicate rows, which of the following SQL statements helps you accomplish this?

- SELECT * FROM orders UNION SELECT * FROM orders_archive
- SELECT * FROM orders INTERSECT SELECT * FROM orders_archive
- SELECT * FROM orders UNION ALL SELECT * FROM orders_archive
- SELECT * FROM orders_archive MINUS SELECT * FROM orders
- SELECT distinct * FROM orders JOIN orders_archive on order.id = orders_archive.id

Unattempted

Answer is SELECT * FROM orders UNION SELECT * FROM orders_archive
UNION and UNION ALL are set operators,
UNION combines the output from both queries but also eliminates the duplicates.
UNION ALL combines the output from both queries.

20. Question

Which of the following python statement can be used to replace the schema name and table name in the query statement?

- table_name = “sales“
schema_name = “bronze“
query = f”select * from schema_name.table_name”
- table_name = “sales“
schema_name = “bronze“
query = “select * from {schema_name}.{table_name}“
- table_name = “sales“
schema_name = “bronze“

query = f“select * from { schema_name}.{table_name}“

table_name = “sales“
schema_name = “bronze“
query = f“select * from + schema_name +“.“+table_name“

Unattempted
Answer is
table_name = “sales”
query = f”select * from {schema_name}.{table_name}”
f strings can be used to format a string. f“ This is string {python variable}“
https://realpython.com/python-f-strings/

21. Question

Which of the following SQL statements can replace python variables in Databricks SQL code, when the notebook is set in SQL mode?
%python
table_name = “sales“
schema_name = “bronze“
%sql
SELECT * FROM _____

SELECT * FROM f{schema_name.table_name}

SELECT * FROM {schem_name.table_name}

SELECT * FROM ${schema_name}.${table_name}

SELECT * FROM schema_name.table_name

Unattempted
The answer is, SELECT * FROM ${schema_name}.${table_name}
%python
table_name = “sales“
schema_name = “bronze“
%sql
SELECT * FROM ${schema_name}.${table_name}
${python variable} -> Python variables in Databricks SQL code

22. Question

A notebook accepts an input parameter that is assigned to a python variable called department and this is an optional parameter to the notebook, you are looking to control the flow of the code using this parameter. you have to check department variable is present then execute the code and if no department value is passed then skip the code execution. How do you achieve this using python?

if department is not None:
#Execute code
else:
pass

if (department is not None)
#Execute code
else
pass

if department is not None:
#Execute code
end:
pass

if department is not None:
#Execute code
then:
pass

if department is None:
#Execute code
else:
pass

Unattempted
The answer is,
if department is not None:
#Execute code
else:
pass

23. Question

Which of the following operations are not supported on a streaming dataset view?
spark.readStream.format(“delta“).table(“sales“).createOrReplaceTempView(“streaming_view“)

SELECT sum(unitssold) FROM streaming_view

SELECT max(unitssold) FROM streaming_view

SELECT id, sum(unitssold) FROM streaming_view GROUP BY id ORDER BY id

SELECT id, count(*) FROM streaming_view GROUP BY id

SELECT * FROM streadming_view ORDER BY id

Unattempted

The answer isSELECT * FROM streadming_view order by id Please Note: Sorting with Group by will work without any issues

see below explanation for each option of the options,

Certain operations are not allowed on streaming data, please see highlighted in bold.

https://spark.apache.org/docs/latest/structured-streaming-programming-guide.html#unsupported-operations

Multiple streaming aggregations (i.e. a chain of aggregations on a streaming DF) are not yet supported on streaming Datasets.

Limit and take the first N rows are not supported on streaming Datasets.

Distinct operations on streaming Datasets are not supported.

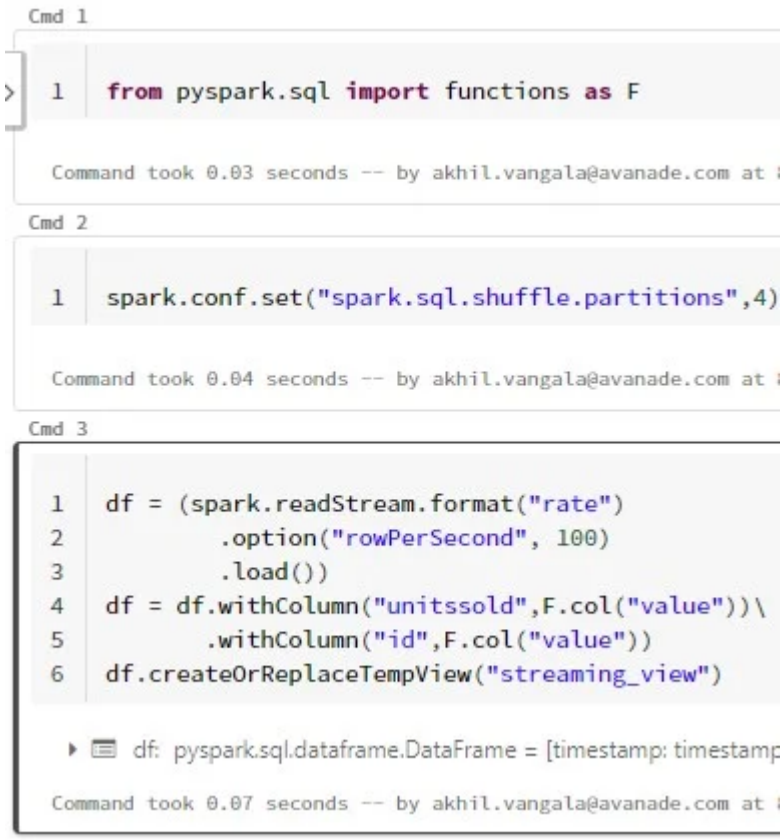
Deduplication operation is not supported after aggregation on a streaming Datasets.

Sorting operations are supported on streaming Datasets only after an aggregation and in Complete Output Mode.

Note: Sorting without aggregation function is not supported.

Here is the sample code to prove this,

Setup test stream



Sum aggregation function has no issues on stream

```
%sql
SELECT sum(unitssold)
FROM streaming_view
```

▸ (1) Spark Jobs

▸ display_query_11 (id: 2eef8b37-4bf8-45e8

	sum(unitssold)
1	66

Max aggregation function has no issues on stream

```
%sql
SELECT max(unitssold)
FROM streaming_view
```

▸ (1) Spark Jobs

▸ display_query_12 (id: c76197eb-d605-4cb5

	max(unitssold)
1	12

Group by with Order by has no issues on stream

```
%sql
SELECT id,count(*)
FROM streaming_view
GROUP BY id
ORDER BY id
```

▸ (1) Spark Jobs

▸ display_query_13 (id: 0e8112ed-bd8e-

	id	count(1)
1	0	1

Group by has no issues on stream

```
%sql
SELECT id,count(*)
FROM streaming_view
GROUP BY id
```

▸ (1) Spark Jobs

▸ display_query_10 (id: c1576177-828e-4cf2

	id	count(1)
1	30	1

Order by without group by fails.

24. Question

Which of the following techniques structured streaming uses to ensure recovery of failures during stream processing?

- Checkpointing and Watermarking
- Write ahead logging and watermarking
- **Checkpointing and write-ahead logging**
- Delta time travel
- The stream will failover to available nodes in the cluster
- Checkpointing and Idempotent sinks

Unattempted

The answer is Checkpointing and write-ahead logging.
Structured Streaming uses checkpointing and write-ahead logs to record the offset range of data being processed during each trigger interval.

25. Question

Which of the statements are incorrect when choosing between lakehouse and Datawarehouse?

- Lakehouse can have special indexes and caching which are optimized for Machine learning
- **Lakehouse cannot serve low query latency with high reliability for BI workloads, only suitable for batch workloads.**
- Lakehouse can be accessed through various API's including but not limited to Python/R/SQL
- Traditional Data warehouses have storage and compute are coupled.

Lakehouse uses standard data formats like Parquet.

Unattempted

The answer is Lakehouse cannot serve low query latency with high reliability for BI workloads, only suitable for batch workloads.

Lakehouse can replace traditional warehouses by leveraging storage and compute optimizations like caching to serve them with low query latency with high reliability.

Focus on comparisons between Spark Cache vs Delta Cache.

https://docs.databricks.com/delta/optimizations/delta-cache.html

What Is a Lakehouse? – The Databricks Blog

26. Question

Which of the statements are correct about lakehouse?

Lakehouse only supports Machine learning workloads and Data warehouses support BI workloads

Lakehouse only supports end-to-end streaming workloads and Data warehouses support Batch workloads

Lakehouse does not support ACID

In Lakehouse Storage and compute are coupled

Lakehouse supports schema enforcement and evolution

Unattempted

The answer is Lakehouse supports schema enforcement and evolution,

Lakehouse using Delta lake can not only enforce a schema on write which is contrary to traditional big data systems that can only enforce a schema on read, it also supports evolving schema over time with the ability to control the evolution.

For example below is the Dataframe writer API and it supports three modes of enforcement and evolution,

Default: Only enforcement, no changes are allowed and any schema drift/evolution will result in failure.

Merge: Flexible, supports enforcement and evolution

New columns are added

Evolves nested columns

Supports evolving data types, like Byte to Short to Integer to Bigint

How to enable:

DF.write.format(“delta“).option(“mergeSchema“, “true“).saveAsTable(“table_name“)

or

spark.databricks.delta.schema.autoMerge = True ## Spark session

Overwrite: No enforcement

Dropping columns

Change string to integer

Rename columns

How to enable:

DF.write.format(“delta“).option(“overwriteSchema“, “True“).saveAsTable(“table_name“)

What Is a Lakehouse? – The Databricks Blog

27. Question

Which of the following are stored in the control pane of Databricks Architecture?

- Job Clusters
- All Purpose Clusters
- Databricks Filesystem
- Databricks Web Application
- Delta tables

Unattempted

The answer is Databricks Web Application

Azure Databricks architecture overview – Azure Databricks | Microsoft Docs

Databricks operates most of its services out of a control plane and a data plane, please note serverless features like SQL Endpoint and DLT compute use shared compute in Control pane.

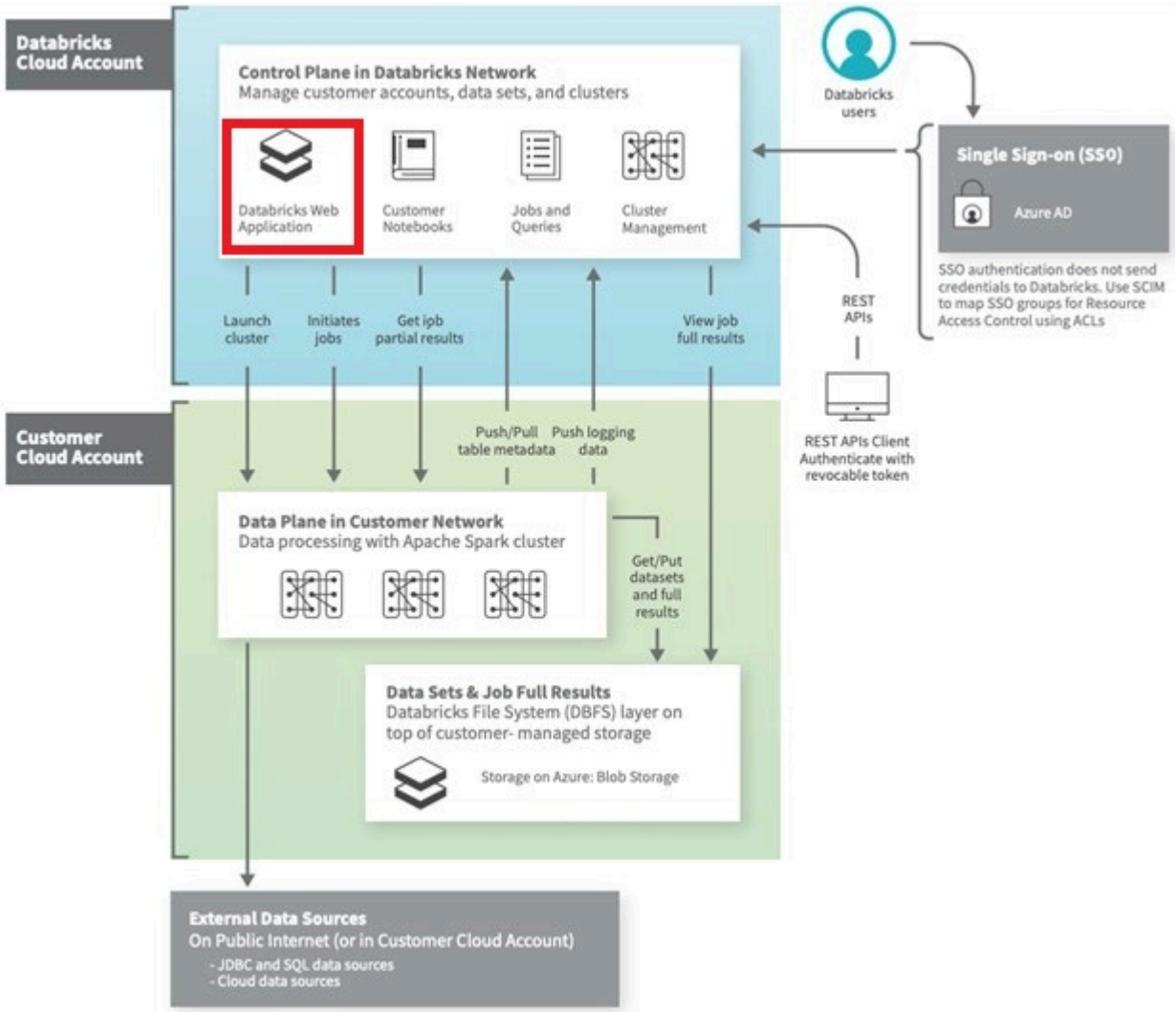
Control Plane: Stored in Databricks Cloud Account

The control plane includes the backend services that Databricks manages in its own Azure account.

Notebook commands and many other workspace configurations are stored in the control plane and encrypted at rest.

Data Plane: Stored in Customer Cloud Account

The data plane is managed by your Azure account and is where your data resides. This is also where data is processed. You can use Azure Databricks connectors so that your clusters can connect to external data sources outside of your Azure account to ingest data or for storage.



28. Question

You have written a notebook to generate a summary data set for reporting, Notebook was scheduled using the job cluster, but you realized it takes 8 minutes to start the cluster, what feature can be used to start the cluster in a timely fashion so your job can run immediatley?

- Setup an additional job to run ahead of the actual job so the cluster is running second job starts
- Use the Databricks cluster pools feature to reduce the startup time
- Use Databricks Premium edition instead of Databricks standard edition
- Pin the cluster in the cluster UI page so it is always available to the jobs
- Disable auto termination so the cluster is always running

Unattempted

Cluster pools allow us to reserve VM's ahead of time, when a new job cluster is created VM are grabbed from the pool. Note: when the VM's are waiting to be used by the cluster only cost incurred is Azure. Databricks run time cost is only billed once VM is allocated to a cluster.
Here is a demo of how to setup a pool and follow some best practices,

29. Question

Which of the following developer operations in the CI/CD can only be implemented through a GIT provider when using Databricks Repos.
Trigger Databricks Repos pull API to update the latest version

- Commit and push code
- Create and edit code
- Create a new branch
- **Pull request and review process**

Unattempted
The answer is Pull request and review process, please note: the question is asking for steps that are being implemented in GIT provider not Databricks Repos.
See below diagram to understand the role of Databricks Repos and Git provider plays when building a CI/CD workdlow.
All the steps highlighted in yellow can be done Databricks Repo, all the steps highlighted in Gray are done in a git provider like Github or Azure Devops.

30. Question

You have noticed the Data scientist team is using the notebook versioning feature with git integration, you have recommended them to switch to using Databricks Repos, which of the below reasons could be the

reason the why the team needs to switch to Databricks Repos.

- Databricks Repos allows multiple users to make changes
- Databricks Repos allows merge and conflict resolution
- Databricks Repos has a built-in version control system
- Databricks Repos automatically saves changes
- **Databricks Repos allow you to add comments and select the changes you want to commit.**

Unattempted

The answer is Databricks Repos allow you to add comments and select the changes you want to commit.

31. Question

Data science team members are using a single cluster to perform data analysis, although cluster size was chosen to handle multiple users and auto-scaling was enabled, the team realized queries are still running slow, what would be the suggested fix for this?

- Setup multiple clusters so each team member has their own cluster
- Disable the auto-scaling feature
- **Use High concurrency mode instead of the standard mode**
- Increase the size of the driver node

Unattempted

The answer is Use High concurrency mode instead of the standard mode,
https://docs.databricks.com/clusters/cluster-config-best-practices.html#cluster-mode
High Concurrency clusters are ideal for groups of users who need to share resources or run ad-hoc jobs.
Databricks recommends enabling autoscaling for High Concurrency clusters.

32. Question

Which of the following SQL commands are used to append rows to an existing delta table?

- APPEND INTO DELTA table_name
- APPEND INTO table_name
- COPY DELTA INTO table_name
- **INSERT INTO table_name**
- UPDATE table_name

Unattempted

The answer is INSERT INTO table_name
Insert adds rows to an existing table, this is very similar to add rows a traditional Database or

Datawarehouse.

33. Question

How are Delt tables stored?

- **A Directory where parquet data files are stored, a sub directory _delta_log where meta data, and the transaction log is stored as JSON files.**
- A Directory where parquet data files are stored, all of the meta data is stored in memory
- A Directory where parquet data files are stored in Data plane, a sub directory _delta_log where meta data, history and log is stored in control pane.
- A Directory where parquet data files are stored, all of the metadata is stored in parquet files
- Data is stored in Data plane and Metadata and delta log are stored in control pane

Unattempted

The answer is A Directory where parquet data files are stored, a sub directory _delta_log where meta data, and the transaction log is stored as JSON files.

34. Question

While investigating a data issue in a Delta table, you wanted to review logs to see when and who updated the table, what is the best way to review this data?

- Review event logs in the Workspace
- Run SQL SHOW HISTORY table_name
- Check Databricks SQL Audit logs
- **Run SQL command DESCRIBE HISTORY table_name**
- Review workspace audit logs

Unattempted

The answer is Run SQL command DESCRIBE HISTORY table_name.

here is the sample data of how DESCRIBE HISTORY table_name looks

+———-+———————-+——+——–+———+——————————+—-+——–+
———+———–+————–+————-+——————–+
|version| timestamp|userId|userName|operation| operationParameters|
job|notebook|clusterId|readVersion|isolationLevel|isBlindAppend| operationMetrics|
+———-+———————-+——+——–+———+——————————+—-+——–+
———+———–+————–+————-+——————–+
| 5|2019-07-29 14:07:47| null| null| DELETE|[predicate -> [“(…|null| null| null| 4| Serializable| false|
[numTotalRows -> …|
| 4|2019-07-29 14:07:41| null| null| UPDATE|[predicate -> (id…|null| null| null| 3| Serializable| false|
[numTotalRows -> …|
| 3|2019-07-29 14:07:29| null| null| DELETE|[predicate -> [“(…|null| null| null| 2| Serializable| false|
[numTotalRows -> …|
| 2|2019-07-29 14:06:56| null| null| UPDATE|[predicate -> (id…|null| null| null| 1| Serializable| false|
[numTotalRows -> …|
| 1|2019-07-29 14:04:31| null| null| DELETE|[predicate -> [“(…|null| null| null| 0| Serializable| false|
[numTotalRows -> …|
| 0|2019-07-29 14:01:40| null| null| WRITE|[mode -> ErrorIfE…|null| null| null| null| Serializable| true|
[numFiles -> 2, n…|
+———-+———————-+——+——–+———+——————————+—-+——–+
———+———–+————–+————-+——————–+

35. Question

While investigating a performance issue, you realized that you have too many small files for a given table, which command are you going to run to fix this issue

- COMPACT table_name
- VACUUM table_name
- MERGE table_name
- SHRINK table_name
- **OPTIMIZE table_name**

Unattempted

The answer is OPTIMIZE table_name,

Optimize compacts small parquet files into a bigger file, by default the size of the files are determined based on the table size at the time of OPTIMIZE, the file size can also be set manually or adjusted based on the workload.

https://docs.databricks.com/delta/optimizations/file-mgmt.html

Tune file size based on Table size

To minimize the need for manual tuning, Databricks automatically tunes the file size of Delta tables based

on the size of the table. Databricks will use smaller file sizes for smaller tables and larger file sizes for larger tables so that the number of files in the table does not grow too large.

36. Question

Create a sales database using the DBFS location ‘dbfs:/mnt/delta/databases/sales.db/‘

- CREATE DATABASE sales FORMAT DELTA LOCATION ‘dbfs:/mnt/delta/databases/sales.db/‘’
- CREATE DATABASE sales USING LOCATION ‘dbfs:/mnt/delta/databases/sales.db/‘
- **CREATE DATABASE sales LOCATION ‘dbfs:/mnt/delta/databases/sales.db/‘**
- The sales database can only be created in Delta lake
- CREATE DELTA DATABASE sales LOCATION ‘dbfs:/mnt/delta/databases/sales.db/‘

Unattempted

The answer is

CREATE DATABASE sales LOCATION ‘dbfs:/mnt/delta/databases/sales.db/‘

Note: with the introduction of the Unity catalog and three-layer namespace usage of SCHEMA and DATABASE is interchangeable

37. Question

What is the type of table created when you issue SQL DDL command CREATE TABLE sales (id int, units int)

- Query fails due to missing location
- Query fails due to missing format

- Managed Delta table
- External Table
- Managed Parquet table

Unattempted

Answer is Managed Delta table

Anytime a table is created without the Location keyword it is considered a managed table, by default all managed tables DELTA tables

Syntax

CREATE TABLE table_name (column column_data_type...)

38. Question

How to determine if a table is a managed table vs external table?

Run IS_MANAGED('table_name') function

- All external tables are stored in data lake, managed tables are stored in DELTA lake
- All managed tables are stored in unity catalog
- Run SQL command DESCRIBE EXTENDED table_name and check type
- A. Run SQL command SHOW TABLES to see the type of the table

Unattempted

The answer is Run SQL command DESCRIBE EXTENDED table_name and check type

Example of External table

Example of managed table

39. Question

Which of the below SQL commands creates a session scoped temporary view?

- **CREATE OR REPLACE TEMPORARY VIEW view_name AS SELECT * FROM table_name**
- CREATE OR REPLACE LOCAL TEMPORARY VIEW view_name AS SELECT * FROM table_name
- CREATE OR REPLACE GLOBAL TEMPORARY VIEW view_name AS SELECT * FROM table_name
- CREATE OR REPLACE VIEW view_name AS SELECT * FROM table_name
- CREATE OR REPLACE LOCAL VIEW view_name AS SELECT * FROM table_name

Unattempted

The answer is
CREATE OR REPLACE TEMPORARY VIEW view_name
AS SELECT * FROM table_name
The default temporary view is session scoped, as soon as the session ends or if a notebook is detached session scoped temporary view is dropped.

40. Question

Drop the customers database and associated tables and data, all of the tables inside the database are managed tables. Which of the following SQL commands will help you accomplish this?

- DROP DATABASE customers FORCE
- **DROP DATABASE customers CASCADE**
- DROP DATABASE customers INCLUDE
- All the tables must be dropped first before dropping database
- DROP DELTA DATABSE customers

Unattempted

The answer is DROP DATABASE customers CASCADE

Drop database with cascade option drops all the tables, since all of the tables inside the database are managed tables we do not need to perform any additional steps to clean the data in the storage.

41. Question

Define an external SQL table by connecting to a local instance of an SQLite database using JDBC

- CREATE TABLE users_jdbc
USING SQLITE
OPTIONS (
url = “jdbc:/sqmple_db“,
dbtable = “users“
)

- CREATE TABLE users_jdbc
USING SQL
URL = {server:“jdbc:/sqmple_db“,dbtable: “users”}

- CREATE TABLE users_jdbc
USING SQL
OPTIONS (
url = “jdbc:sqlite:/sqmple_db“,
dbtable = “users“
)

- CREATE TABLE users_jdbc
USING org.apache.spark.sql.jdbc.sqlite
OPTIONS (
url = “jdbc:/sqmple_db“,
dbtable = “users“
)

- **CREATE TABLE users_jdbc**
USING org.apache.spark.sql.jdbc
OPTIONS (
url = “jdbc:sqlite:/sqmple_db“,
dbtable = “users“
)

Unattempted

The answer is,

CREATE TABLE users_jdbc
USING org.apache.spark.sql.jdbc
OPTIONS (
url = “jdbc:sqlite:/sqmple_db“,

dbtable = “users“
)
Databricks runtime currently supports connecting to a few flavors of SQL Database including SQL Server, My SQL, SQL Lite and Snowflake using JDBC.
CREATE TABLE
USING org.apache.spark.sql.jdbc or JDBC
OPTIONS (
url = “jdbc::///:“,
dbtable “ = .atable“,
user = ““,
password = ““
)
For more detailed documentation
SQL databases using JDBC – Azure Databricks | Microsoft Docs

42. Question

When defining external tables using formats CSV, JSON, TEXT, BINARY any query on the external tables caches the data and location for performance reasons, so within a given spark session any new files that may have arrived will not be available after the initial query. How can we address this limitation?

- UNCACHE TABLE table_name
- CACHE TABLE table_name
- **REFRESH TABLE table_name**
- BROADCAST TABLE table_name
- CLEAR CACH table_name

Unattempted
The answer is REFRESH TABLE table_name
REFRESH TABLE table_name will force Spark to refresh the availability of external files and any changes. When spark queries an external table it caches the files associated with it, so that way if the table is queried again it can use the cached files so it does not have to retrieve them again from cloud object storage, but the drawback here is that if new files are available Spark does not know until the Refresh command is ran.

43. Question

Which of the following table constraints that can be enforced on Delta lake tables are supported?

- Primary key, foreign key, Not Null, Check Constraints
- Primary key, Not Null, Check Constraints
- Default, Not Null, Check Constraints
- **Not Null, Check Constraints**

Unique, Not Null, Check Constraints

Unattempted

The answer is Not Null, Check Constraints

https://docs.microsoft.com/en-us/azure/databricks/delta/delta-constraints

CREATE TABLE events(id LONG,

date STRING,

location STRING,

description STRING

) USING DELTA;

ALTER TABLE events CHANGE COLUMN id SET NOT NULL;

ALTER TABLE events ADD CONSTRAINT dateWithinRange CHECK (date > '1900-01-01');

Note: Databricks as of DBR 11.1 added support for Primary Key and Foreign Key when Unity Catalog is enabled but this is for information purposes only these are not actually enforced. You may ask then why are we defining these if they are not enforced, so especially these information constraints are very helpful if you have a BI tool that can benefit from knowing the relationship between the tables, so it will be easy when creating reports/dashboards or understanding the data model when using any Data modeling tool.

Primary and Foreign Key

44. Question

The data engineering team is looking to add a new column to the table, but the QA team would like to test the change before implementing in production, which of the below options allow you to quickly copy the table from Prod to the QA environment, modify and run the tests?

DEEP CLONE

SHADOW CLONE

ZERO COPY CLONE

SHALLOW CLONE

METADATA CLONE

Unattempted

The answer is SHALLOW CLONE

SHALLOW CLONE If you wish to create a copy of a table quickly to test out applying changes without the risk of modifying the current table, SHALLOW CLONE can be a good option. Shallow clones just copy the

Delta transaction logs, meaning that the data doesn't move so it can be very quick.
CREATE OR REPLACE TABLE {new_table_name} SHALLOW CLONE {source_table_name}|[LOCATION path]
DEEP CLONE fully copies data and metadata from a source table to a target. This copy occurs incrementally, so executing this command again can sync changes from the source to the target location. It copies all of the data and transaction logs this can take a long time based on the size of the table.
CREATE OR REPLACE TABLE {new_table_name} DEEP CLONE {source_table_name}|[LOCATION path]

45. Question

Sales team is looking to get a report on a measure number of units sold by date, below is the schema. Fill in the blank with the appropriate array function.
Table orders: orderDate DATE, orderIds ARRAY
Table orderDetail: orderId INT, unitsSold INT, salesAmt DOUBLE

SELECT orderDate, SUM(unitsSold)
FROM orderDetail od
JOIN (select orderDate, _____(orderIds) as orderId FROM orders) o
ON o.orderId = od.orderId
GROUP BY orderDate

- FLATTEN
- EXTEND
- **EXPLODE**
- EXTRACT
- ARRAY_FLATTEN

Unattempted
The answer is EXPLODE,
explode is table-valued function, takes an array or map and returns a row for each element in the array.
For below table

select explode(orderIds) orderId, orderdate from orders

This above would return a result something like below, creates a row for each element in the array

OrderDate	orderId
10-10-2021	100
10-10-2021	101
10-10-2021	102

..

46. Question

You are asked to write a python function that can read data from a delta table and return the DataFrame, which of the following is correct?

- Python function cannot return a DataFrame
- Write SQL UDF to return a DataFrame
- Write SQL UDF that can return tabular data
- Python function will result in out of memory error due to data volume
- **Python function can return a DataFrame**

Unattempted

The answer is Python function can return a DataFrame

The function would something like this,

```
get_source_dataframe(tablename):
df = spark.read.table(tablename)
return df
df = get_source_dataframe('test_table')
```

since there is no action spark returns a Dataframe and assigns to df python variable

47. Question

What is the output of the below function when executed with input parameters 1, 3 :

```
def check_input(x,y):
if x < y:
x= x+1
if x x= x+1
if x x = x+1
return x
check_input(1,3)
```

- 1
- 2
- **3**
- 4

5

Unattempted

48. Question

Which of the following SQL statements can replace a python variable, when the notebook is set in SQL mode

table_name = “sales“

schema_name = “bronze“

- spark.sql(f“SELECT * FROM f{schema_name.table_name}“)
- spark.sql(f“SELECT * FROM {schem_name.table_name}“)
- spark.sql(f“SELECT * FROM ${schema_name}.${table_name}“)
- **spark.sql(f“SELECT * FROM {schema_name}.{table_name}“)**
- spark.sql(“SELECT * FROM schema_name.table_name“)

Unattempted

The answer is spark.sql(f“SELECT * FROM {schema_name}.{table_name}“)

49. Question

When writing streaming data, Spark’s structured stream supports the below write modes

- Append, Delta, Complete
- Delta, Complete, Continuous
- **Append, Complete, Update**
- Complete, Incremental, Update
- Append, overwrite, Continuous

Unattempted

The answer is Append, Complete, Update

Append mode (default) – This is the default mode, where only the new rows added to the Result Table since the last trigger will be outputted to the sink. This is supported for only those queries where rows added to the Result Table is never going to change. Hence, this mode guarantees that each row will be output only once (assuming fault-tolerant sink). For example, queries with

only select, where, map, flatMap, filter, join, etc. will support Append mode.

Complete mode – The whole Result Table will be outputted to the sink after every trigger. This is supported for aggregation queries.

Update mode – (Available since Spark 2.1.1) Only the rows in the Result Table that were updated since the last trigger will be outputted to the sink. More information to be added in future releases.

50. Question

When using the complete mode to write stream data, how does it impact the target table?

- Entire stream waits for complete data to write
- Stream must complete to write the data
- Target table cannot be updated while stream is pending
- **Target table is overwritten for each batch**
- Delta commits transaction once the stream is stopped

Unattempted

The answer is Target table is overwritten for each batch

Complete mode – The whole Result Table will be outputted to the sink after every trigger. This is supported for aggregation queries

51. Question

At the end of the inventory process a file gets uploaded to the cloud object storage, you are asked to build a process to ingest data which of the following method can be used to ingest the data incrementally, the schema of the file is expected to change overtime ingestion process should be able to handle these changes automatically. Below is the auto loader command to load the data, fill in the blanks for successful execution of the below code.

```
spark.readStream
.format("cloudfiles")
.option("cloudfiles.format","csv)
.option("_______", 'dbfs:/location/checkpoint/')
.load(data_source)
.writeStream
.option("_______",' dbfs:/location/checkpoint/')
.option("mergeSchema", "true")
.table(table_name))
```

- checkpointlocation, schemalocation
- checkpointlocation, cloudfiles.schemalocation
- schemalocation, checkpointlocation
- **cloudfiles.schemalocation, checkpointlocation**
- cloudfiles.schemalocation, cloudfiles.checkpointlocation

Unattempted

The answer is cloudfiles.schemalocation, checkpointlocation

When reading the data cloudfiles.schemalocation is used to store the inferred schema of the incoming data.

When writing a stream to recover from failures checkpointlocation is used to store the offset of the byte that was most recently processed.

52. Question

When working with AUTO LOADER you noticed that most of the columns that were inferred as part of loading are string data types including columns that were supposed to be integers, how can we fix this?

- Provide the schema of the source table in the cloudfiles.schemalocation
- Provide the schema of the target table in the cloudfiles.schemalocation
- **Provide schema hints**
- Update the checkpoint location
- Correct the incoming data by explicitly casting the data types

Unattempted

The answer is, Provide schema hints.

```
spark.readStream \
.format("cloudFiles") \
.option("cloudFiles.format", "csv") \
.option("header", "true") \
.option("cloudFiles.schemaLocation", schema_location) \
.option("cloudFiles.schemaHints", "id int, description string")
.load(raw_data_location)
.writeStream \
.option("checkpointLocation", checkpoint_location) \
.start(target_delta_table_location)
.option("cloudFiles.schemaHints", "id int, description string")
# Here we are providing a hint that id column is int and the description is a string
```

When cloudfiles.schemalocation is used to store the output of the schema inference during the load process, with schema hints you can enforce data types for known columns ahead of time.

53. Question

You have configured AUTO LOADER to process incoming IOT data from cloud object storage every 15 mins, recently a change was made to the notebook code to update the processing logic but the team later realized that the notebook was failing for the last 24 hours, what steps team needs to take to reprocess the data that was not loaded after the notebook was corrected?

Move the files that were not processed to another location and manually copy the files into the ingestion path to reprocess them

- Enable back_fill = TRUE to reprocess the data
- Delete the checkpoint folder and run the autoloader again
- **Autoloader automatically re-processes data that was not loaded**

Manually re-load the data

Unattempted

The answer is,

Autoloader automatically re-processes data that was not loaded using the checkpoint.

54. Question

Which of the following Structured Streaming queries is performing a hop from a bronze table to a Silver table?

(spark.table(“sales“).groupBy(“store“)
.agg(sum(“sales“)).writeStream
.option(“checkpointLocation“,checkpointPath)
.outputMode(“complete“)
.table(“aggregatedSales“))

(spark.table(“sales“).agg(sum(“sales“),sum(“units“))
.writeStream
.option(“checkpointLocation“,checkpointPath)
.outputMode(“complete“)
.table(“aggregatedSales“))

**(spark.table(“sales“)
.withColumn(“avgPrice“, col(“sales“) / col(“units“))
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“cleanedSales“))**

(spark.readStream.load(rawSalesLocation)
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“uncleanedSales“))

(spark.read.load(rawSalesLocation)
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“uncleanedSales“))

Unattempted

The question is asking to identify a structured streaming command that is moving data from bronze to silver.

The answer is

(spark.table(“sales“)

.withColumn(“avgPrice“, col(“sales“) / col(“units“))
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“cleanedSales“))

We are preserving the grain of incoming data and enriching the data by adding avg price, the other options listed use aggregations which are mostly performed on top of the silver to move data to Gold.

Exam focus: Please review the below image and understand the role of each layer(bronze, silver, gold) in medallion architecture, you will see varying questions targeting each layer and its purpose.

55. Question

Which of the following Structured Streaming queries successfully performs a hop from a Silver to Gold table?

(spark.table(“sales“)
.groupBy(“store“)
.agg(sum(“sales“))
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“complete“)
.table(“aggregatedSales“))

(spark.table(“sales“)
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“complete“)
.table(“sales“))

(spark.table(“sales“)
.withColumn(“avgPrice“, col(“sales“) / col(“units“))
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“cleanedSales“))

(spark.readStream.load(rawSalesLocation)
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“uncleanedSales“))

(spark.read.load(rawSalesLocation)
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“append“)
.table(“uncleanedSales“))

Unattempted

The answer is
(spark.table(“sales“)
.groupBy(“store“)
.agg(sum(“sales“))
.writeStream
.option(“checkpointLocation“, checkpointPath)
.outputMode(“complete“)
.table(“aggregatedSales“))
The gold layer is normally used to store aggregated data
Review the below link for more info,
Medallion Architecture – Databricks
Gold Layer:
1. Powers Ml applications, reporting, dashboards, ad hoc analytics
2. Refined views of data, typically with aggregations
3. Reduces strain on production systems
4. Optimizes query performance for business-critical data
Exam focus: Please review the below image and understand the role of each layer(bronze, silver, gold) in medallion architecture, you will see varying questions targeting each layer and its purpose.

56. Question

Which of the following Auto loader structured streaming commands successfully performs a hop from the landing area into Bronze?

- spark\
.readStream\
.format(“csv“)\
.option(“cloudFiles.schemaLocation“, checkpoint_directory)\
.load(“landing“)\
.writeStream.option(“checkpointLocation“, checkpoint_directory)\
.table(raw)

- **spark**
.readStream
.format(“cloudFiles“)
.option(“cloudFiles.format“,“csv“)
.option(“cloudFiles.schemaLocation“, checkpoint_directory)
.load(“landing“)
.writeStream.option(“checkpointLocation“, checkpoint_directory)
.table(raw)

- spark\
.read\
.format(“cloudFiles“)\
.option(“cloudFiles.format“,”csv”)\
.option(“cloudFiles.schemaLocation“, checkpoint_directory)\
.load(“landing“)\

.writeStream.option(“checkpointLocation“, checkpoint_directory)\
.table(raw)

spark\
.readStream\
.load(rawSalesLocation)\
.writeStream \
.option(“checkpointLocation“, checkpointPath).outputMode(“append“)\
.table(“uncleanedSales“)

spark\
.read\
.load(rawSalesLocation) \
.writeStream\
.option(“checkpointLocation“, checkpointPath) \
.outputMode(“append“)\
.table(“uncleanedSales“)

Unattempted

The answer is
spark\
.readStream\
.format(“cloudFiles“) \# use Auto loader
.option(“cloudFiles.format“,“csv“) \ # csv format files
.option(“cloudFiles.schemaLocation“, checkpoint_directory)\
.load(‘landing‘)\
.writeStream.option(“checkpointLocation“, checkpoint_directory)\
.table(raw)
Note: if you chose the below option which is incorrect because it does not have readStream
spark.read.format(“cloudFiles“)
.option(“cloudFiles.format“,”csv”)

..
..
Exam focus: Please review the below image and understand the role of each layer(bronze, silver, gold) in medallion architecture, you will see varying questions targeting each layer and its purpose.

57. Question

A DELTA LIVE TABLE pipelines can be scheduled to run in two different modes, what are these two different modes?

- Triggered, Incremental
- Once, Continuous
- **Triggered, Continuous**
- Once, Incremental
- Continuous, Incremental

Unattempted

The answer is Triggered, Continuous

https://docs.microsoft.com/en-us/azure/databricks/data-engineering/delta-live-tables/delta-live-tables-concepts#—continuous-and-triggered-pipelines

Triggered pipelines update each table with whatever data is currently available and then stop the cluster running the pipeline. Delta Live Tables automatically analyzes the dependencies between your tables and starts by computing those that read from external sources. Tables within the pipeline are updated after their dependent data sources have been updated.

Continuous pipelines update tables continuously as input data changes. Once an update is started, it continues to run until manually stopped. Continuous pipelines require an always-running cluster but ensure that downstream consumers have the most up-to-date data.

58. Question

Your team member is trying to set up a delta pipeline and build a second gold table to the same pipeline with aggregated metrics based on an existing Delta Live table called sales_orders_cleaned but he is facing a problem in starting the pipeline, the pipeline is failing to state it cannot find the table sales_orders_cleaned, you are asked to identify and fix the problem.

```
CREATE LIVE TABLE sales_order_in_chicago
AS
SELECT order_date, city, sum(price) as sales,
FROM sales_orders_cleaned
WHERE city = 'Chicago')
GROUP BY order_date, city
```

- Use STREAMING LIVE instead of LIVE table
- Delta live table can be used in a group by clause
- Delta live tables pipeline can only have one table
- **Sales_orders_cleaned table is missing schema name LIVE**
- The pipeline needs to be deployed so the first table is created before we add a second table

Unattempted

The answer is, Sales_orders_cleaned table is missing schema name LIVE

Every Delta live table should have schema LIVE

Here is the correct syntax,

```
CREATE LIVE TABLE sales_order_in_chicago
AS
SELECT order_date, city, sum(price) as sales,
FROM LIVE.sales_orders_cleaned
WHERE city = 'Chicago')
GROUP BY order_date, city
```

59. Question

Which of the following type of tasks cannot setup through a job?

- Notebook
- DELTA LIVE PIPELINE
- Spark Submit
- Python
- **Databricks SQL Dashboard refresh**

Unattempted

60. Question

Which of the following approaches can the data engineer use to obtain a version-controllable configuration of the Job's schedule and configuration?

- They can link the Job to notebooks that are a part of a Databricks Repo.
- They can submit the Job once on a Job cluster.
- **They can download the JSON equivalent of the job from the Job's page.**
- They can submit the Job once on an all-purpose cluster.
- They can download the XML description of the Job from the Job's page

Unattempted

61. Question

What steps need to be taken to set up a DELTA LIVE PIPELINE as a job using the workspace UI?
DELTA LIVE TABLES do not support job cluster

- **Select Workflows UI and Delta live tables tab, under task type select Delta live tables pipeline and select the notebook**
- Select Workflows UI and Delta live tables tab, under task type select Delta live tables pipeline and select the pipeline JSON file
- Use Pipeline creation UI, select a new pipeline and job cluster

Unattempted

The answer is,

Select Workflows UI and Delta live tables tab, under task type select Delta live tables pipeline and select the notebook.

Create a pipeline

To create a new pipeline using the Delta Live Tables notebook:

1. Click  Workflows in the sidebar, click the Delta Live Tables tab, and click Create Pipeline.
2. Give  the pipeline a name and click to select a notebook.
3. Optionally enter a storage location for output data from the pipeline. The system uses a default location if you leave Storage Location empty.
4. Select Triggered for Pipeline Mode.
5. Click Create.

The system displays the Pipeline Details page after you click Create. You can also access your pipeline by clicking the pipeline name in the Delta Live Tables tab.

62. Question

Data engineering team has provided 10 queries and asked Data Analyst team to build a dashboard and refresh the data every day at 8 AM, identify the best approach to set up data refresh for this dashaboard?

Each query requires a separate task and setup 10 tasks under a single job to run at 8 AM to refresh the dashboard

The entire dashboard with 10 queries can be refreshed at once, single schedule needs to be set up to refresh at 8 AM.

Setup JOB with linear dependency to all load all 10 queries into a table so the dashboard can be refreshed at once.

A dashboard can only refresh one query at a time, 10 schedules to set up the refresh.

Use Incremental refresh to run at 8 AM every day.

Unattempted

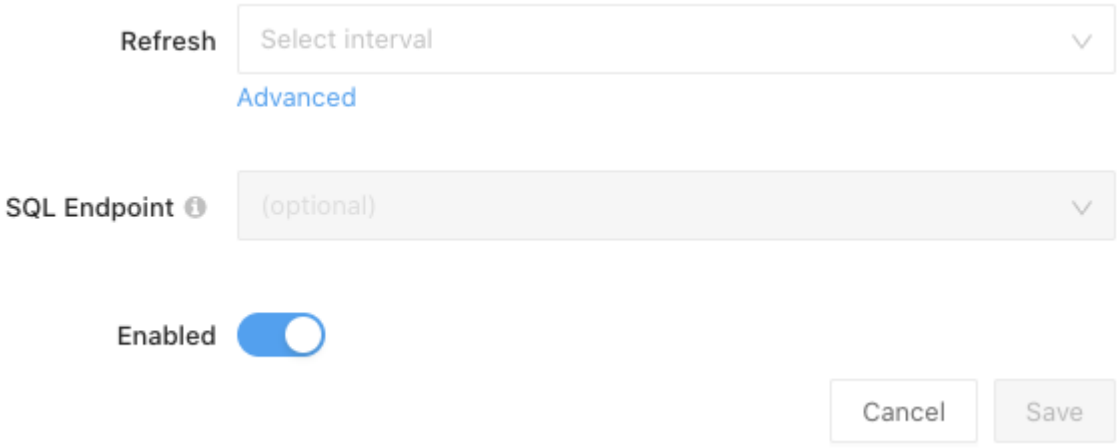
The answer is,

The entire dashboard with 10 queries can be refreshed at once, single schedule needs to be set up to refresh at 8 AM.

Automatically refresh a dashboard

A dashboard's owner and users with the Can Edit permission can configure a dashboard to automatically refresh on a schedule. To automatically refresh a dashboard:

1. Click the Schedule button at the top right of the dashboard. The scheduling dialog appears.



2. In the Refresh every drop-down, select a period.
3. In the SQL Warehouse drop-down, optionally select a SQL warehouse to use for all the queries. If you don't select a warehouse, the queries execute on the last used SQL warehouse.
4. Next to Subscribers, optionally enter a list of email addresses to notify when the dashboard is automatically updated.

Each email address you enter must be associated with a Azure Databricks account or configured as an alert destination.

5. Click Save. The Schedule button label changes to Scheduled.

63. Question

The data engineering team is using a SQL query to review data completeness every day to monitor the ETL job, and query output is being used in multiple dashboards which of the following approaches can be used to set up a schedule and automate this process?

They can schedule the query to run every day from the Jobs UI.

They can schedule the query to refresh every day from the query's page in Databricks SQL

They can schedule the query to run every 12 hours from the Jobs UI.

They can schedule the query to refresh every day from the SQL endpoint's page in Databricks SQL.

They can schedule the query to refresh every 12 hours from the SQL endpoint's page in Databricks SQL

Unattempted

The answer is They can schedule the query to refresh every 12 hours from the SQL endpoint's page in Databricks SQL,

The query pane view in Databricks SQL workspace provides the ability to add or edit and schedule individual queries to run.

You can use scheduled query executions to keep your dashboards updated or to enable routine alerts. By default, your queries do not have a schedule.

Note

If your query is used by an alert, the alert runs on its own refresh schedule and does not use the query schedule.

To set the schedule:

1. Click the query info tab.

2. Click the link to the right of Refresh Schedule to open a picker with schedule intervals.

3. Set the schedule.

The picker scrolls and allows you to choose:

An interval: 1-30 minutes, 1-12 hours, 1 or 30 days, 1 or 2 weeks

A time. The time selector displays in the picker only when the interval is greater than 1 day and the day selection is greater than 1 week. When you schedule a specific time, Databricks SQL takes input in your computer's timezone and converts it to UTC. If you want a query to run at a certain time in UTC, you must adjust the picker by your local offset. For example, if you want a query to execute at 00:00 UTC each day, but your current timezone is PDT (UTC-7), you should select 17:00 in the picker:

64. Question

A data engineer is using a Databricks SQL query to monitor the performance of an ELT job. The ELT job is triggered by a specific number of input records being ready to process. The Databricks SQL query returns the number of minutes since the job's most recent runtime. Which of the following approaches can enable the data engineering team to be notified if the ELT job has not been run in an hour?

They can set up an Alert for the accompanying dashboard to notify them if the returned value is greater than 60.

They can set up an Alert for the query to notify when the ELT job fails.

They can set up an Alert for the accompanying dashboard to notify when it has not refreshed in 60 minutes.

They can set up an Alert for the query to notify them if the returned value is greater than 60.

This type of alert is not possible in Databricks

Unattempted

The answer is, They can set up an Alert for the query to notify them if the returned value is greater than 60.

The important thing to note here is that alert can only be setup on query not on the dashboard, query can return a value, which is used if alert can be triggered.

65. Question

Which of the following programming languages can be used to build a Databricks SQL dashboard?

Python

Scala

SQL

R

All of the above

Unattempted

The answer is SQL

Use Page numbers below to navigate to other practice tests

Skillcertpro

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